Revising science-relevant beliefs:
A meta-analysis of misinformation correction and misconception update

PRISMA-P Protocol

We prepare this protocol by following Moreau and Beau (2022)'s recommendations for open science practices to meta-analysis. Items in this protocol correspond to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols Statement (PRISMA-P; Moher et al., 2016).

SECTION 1: Administrative Information

Title

Item 1a. *Identify the report as a protocol of a systematic review.*

This is a preregistration protocol of a meta-analysis.

Item 1b. *If the protocol is for an update of a previous systematic review, identify as such.*

This protocol is for an update of a previous meta-analysis done on 100% of the data in March 2022. The updates include adding database search results from January 1, 2022, to August 31, 2022, and moderators in the analyses.

Registration

Item 2. *If registered, provide the name of the registry (e.g., PROSPERO) and registration number.*

N/A

Authors

Item 3a. *Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author.*

Man-pui Sally Chan, University of Pennsylvania, sallycmp@upenn.edu, 3620, Walnut St, Philadelphia, PA 19104. Dolores Albarracin, University of Pennsylvania, dalba@upenn.edu.

Item 3b. *Describe contributions of protocol authors and identify the guarantor of the review.*

Man-pui Sally Chan: Conceptualization, Data curation, Project administration, Software, Validation, Visualization, Writing protocol – original draft. Dolores Albarracin: Conceptualization, Data curation, Methodology, Writing protocol – review & editing.

Amendments

Item 4. *If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments.*

N/A

Support

Item 5a. *Indicate sources of financial or other support for the review.*

No.

Item 5b. *Provide name for the review funder and/or sponsor.*

N/A

Item 5c. *Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol.*

N/A

SECTION 2: Introduction

Rationale

Item 6. *Describe the rationale for the review in the context of what is already known.*

Effectively correcting scientifically relevant information, including information about a false research claim or conclusion based on a scientific measurement procedure, is important. Six prior meta-analyses have examined the persistence of misinformation in news and reports (Chan, Jones, Hall Jamieson, et al., 2017; Janmohamed et al., 2022; Walter et al., 2020, 2021; Walter & Murphy, 2018; Walter & Tukachinsky, 2020). Specifically, Chan et al. (2017) meta-analyzed eight reports (*k* of effect sizes = 52) of research that used fictitious social and political news as experimental materials. Two other meta-analyses concentrated on social and political news, such as restaurant rumors and news about political events (Walter et al., 2020; Walter & Tukachinsky, 2020). However, these meta-analyses did not concentrate on science-relevant misinformation. The only three that did were conducted by Walter and colleagues (Janmohamed et al., 2022; Walter et al., 2021; Walter & Murphy, 2018). One synthesized six effect sizes on the impact of environmental misinformation corrections for naturally occurring misinformation about climate change (Walter & Murphy, 2018). Another synthesized five effect sizes on interventions to correct vaping misinformation (Janmohamed et al., 2022). The other synthesized nineteen experiments about the impact of a source reliability rating on corrections of misleading health news on social media (e.g., a claim about universities merging their healthcare systems with Planned Parenthood) (Walter et al., 2021). However, these meta-analyses considered a small number of effect sizes (i.e., $k = 5 - 24$) in a limited set of science-relevant studies, and neither examined multiple moderating roles of the misinformation, the correction, and the recipient of the misinformation on corrections. A comprehensive analysis of the factors that affect the correction of initial information coming from scientific and academic sources is lacking despite the existence of experimental evidence in the area of correcting science-relevant beliefs.

Objectives

Item 7. *Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO).*

We will synthesize a large body of experimental evidence comprising published experiments, working papers, dissertation and thesis, and unpublished datasets. In particular, we attempt to examine the moderating effects of factors related to the misconception or misinformation (i.e., negativity, domain, polarization, and veracity), the correction (i.e., detailed correction and in-person correction), and the recipient (i.e., attitude congruence, likely prior exposure, geographic location of the participants, and data collection).

SECTION 3: Methods

Eligibility criteria

Item 8. *Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review.*

We include studies about (a) science communication. That is, we include reports presenting information based on a scientific measurement procedure and scientific evidence allegedly published in a scientific journal. We include studies (b) empirically measured participants' beliefs in or attitudes about the presented misinformation. However, studies that included outcome measures of misinformation sharing (intentions) (Arechar et al., 2022; Jahanbakhsh et al., 2021; Pennycook et al., 2020, 2021; Sirlin et al., 2021), the quality judgment of news sources (Pennycook & Rand, 2019), self-efficacy, openness to the messages (Gesser-Edelsburg et al., 2018), and whether respondents responded to/ignored the corrections were excluded (Mosleh et al., 2022). Included studies (c) had a control or baseline group exposed to either no message, a neutral message, or an unrelated message; and (d) used either new or existing information. Studies that used existing and novel topics were both eligible. However, studies that described the initial information as hypothetical or uncertain (e.g., Koller, 1993) or as a correction of another piece of information (e.g., Greitemeyer & Sagioglou, 2015) were excluded. The selected studies that introduced misinformation (e) must initially assert the misinformation to be true, and all studies introduced a correction for the misinformation regardless of whether the misinformation was introduced in the experiment or known to be familiar to the participants (e.g., Calabrese & Albarracin, n.d.).

Information sources

Item 9. *Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage.*

We use specific keywords with wildcards, performing a combined search of the following seven online databases: (a) PsycInfo, (b) Google Scholar, (c) MEDLINE, (d) PubMed, (e) ProQuest Dissertations and Theses Abstracts and Indexes: Social Sciences, (f) Communication Source, and (g) Web of Science Social Sciences Citation Index. We completed the searches in December 2021, screened the search results in February 2022, and finished coding the moderators in March 2022. In this protocol, we will update the included the flow diagram figure (see Figure 1) in the preregistration. We decide not to use the Scopus database because of substantial overlapping with PsycInfo.

Search strategy

Item 10.

We pair a series of keywords (i.e., misinformation OR misbelief* OR false information OR belief perseverance OR continued influence) with two other series of keywords (i.e., [retract* OR correct*] and [label* OR tag* OR flag*]). Table 1 shows the search syntax for each database.

Study records

Item 11a. *Describe the mechanism(s) that will be used to manage records and data throughout the review.*

We will manage all searched results and data in .ris, .txt, and .xlsx formats. We first download search results as .ris files and import them into Mendeley, reference management software, to check duplicates. Next, we export the unique records into .txt files for screening. We then input the included records into an excel file for coding data (see TEMPLATE IncludedResults-StudyNumber.xlsx).

Item 11b. State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis).

Two authors coded for five moderators in early 2022, including the variables of interest: (a) whether the misconception or misinformation was about negative or neutral topics, (b) the level of detail of the correction messages, (c) the use of in-person in correction, (d) the attitudinal congruence of the correction, and (e) the likelihood of prior exposure to the topic (Ecker, 2022; Ecker et al., 2022; Gawronski et al., 2022; Hart et al., 2009; Lewandowsky et al., 2012, 2020; Lord et al., 1979; Seifert, 2002; van der Linden et al., 2017). The coding of all variables reached adequate agreement in all cases (kappa: M = .90, SD = .13 and icc: M = .89, SD = .15). In late August 2022, two authors discussed a new coding scheme of additional five variables, including (f) whether the misconception or misinformation was about politics, health, environment, and others, (g) whether the topic is polarized, (h) a lack of scientific support of the veracity of the claim, (i) the geographical location of the participants, and (j) whether the experiment was carried out at laboratory or online. We will use a subset of the codes to establish the reliability of new variables. Once the reliability is established, one coder will complete the coding of new variables independently.

Item 11c. *Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators.*

The coder will independently extract data from reports using either the between-subjects or within-subjects templates. (see TEMPLATE between_effect-sizes_2022v2.xlsx and TEMPLATE within_effect-sizes_2022v2.xlsx). We will then combine records of each study into a master data file.

Data items

Item 12. *List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications.*

A. Moderators

Topic of misconception or misinformation. We coded whether the misinformation topic was negative or neutral. (negative = 2; neutral = 1).

Level of detail of the correction messages. We coded whether the correction provided detailed information or simply labeled the initial information as incorrect (detailed = 2; succinct = 1).

In-person correction. We coded whether the correction was given in person or not (yes = 1; no = -1).

Attitudinal congruence of the correction. This variable captured whether participants had any pre-existing attitudes relative to the position advocated in the correction message (congruent information = 1; either inconsistent or nonpartisan information = 0).

Likely prior exposure to the topic. We coded for whether the topic used in the experiment had circulated in the real world (likely with exposure = 1; likely without prior exposure = -1).

Domain of the misconception or misinformation. We will code whether the misconception or misinformation was about politics, health, environment, and others (politics = 1; health = 2; environment = 3; others = 4).

Polarization of the topic. We will code whether the topic is associated with disagreement between opposing groups in the country where the experiment was carried out (yes = 1; no = -1).

Veracity of the misconception or misinformation. We will code whether there is a lack of scientific support of the veracity of the claim (no scientific support = 1; with scientific support = -1). For example, the alleged link between Zika virus vaccines and epilepsy was never true (1) whereas preliminary evidence showed that hydroxychloroquine was shown to be effective against COVID-19 (-1).

Geographical location of the participants. We will code whether participants were self-reported as from the U.S. or other countries (U.S. = 1; other countries = 2).

Data collection. We will code whether the experiment was carried out in the lab or online (lab = 1; online = 2).

B. Study characteristics

Source type. We will code the report's publication category (published article = 1; working paper = 5; dissertation/thesis = 6; unpublished data = 7).

Mean age. We will code the mean age of participants who participated in the experiment regardless of conditions.

Percentage of females. We will code the percentages of female participants who participated in the experiment regardless of conditions.

Percentage of males. We will code the percentages of male participants who participated in the experiment regardless of conditions.

Outcomes and prioritisation

Item 13. *List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale.*

N/A

Risk of bias in individual studies

Item 14. *Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis.*

We will adopt the diagnostic procedures proposed by Viechtbauer and Cheung (2010) to detect influential cases.

Data synthesis

Item 15a. *Describe criteria under which study data will be quantitatively synthesized.*

We will use the results of influential analyses to determine outliers and repeat the analyses with and without outliers.

Item 15b. *If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I 2 , Kendall's tau).*

Debunking effects involve both correction and the reverse of belief perseverance or misinformation persistence, the analyses will utilize RVE to account for the statistical dependence between the two effect sizes. We will perform simple correlation analyses to estimate the correlation parameter. We will then analyze the mean effect sizes using robust variance estimation (RVE) methods (Fisher & Tipton, 2015; Hedges et al., 2010; Sidik & Jonkman, 2006). We will also calculate the I^2 statistic (i.e., the percentage of total variation across experimental conditions due to random heterogeneity), which controls for k and indicates the percentage of total variation across experimental conditions that is due to true heterogeneity rather than sampling error (Higgins et al., 2003; Higgins & Thompson, 2002). Furthermore, we will perform meta-regression analyses of our debunking effect sizes with the moderators.

We will use Hedges' *d* as the metric for effect size estimation in our meta-analysis. Coders will first decide whether the report was in a between- or within-subjects design and then select corresponding means and standard deviations from different groups or conditions to compute Hedges' *d* in accordance with the formulas outlined by Borenstein et al. (2009). All *d* statistics are in the same normalized units regardless of whether they derive from between- or within-subjects designs (Borenstein et al., 2009, p. 26).

Item 15c. *Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression).*

We will conduct sensitivity analyses by applying imputation techniques to estimate missing misinformation effect sizes and performing Bayesian analyses if there are null effects of the moderator analyses.

Item 15d. *If quantitative synthesis is not appropriate, describe the type of summary planned.*

N/A

Meta-bias(es)

Item 16. *Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies).*

We will follow McShane et al. (2016)'s recommendation for assessing publication bias. We will perform seven bias tests to assess publication/inclusion biases for effect sizes with and without the outlier. Specifically, we will use R packages: *metaphor* (Viechtbauer, 2010), and *weightr* (Coburn & Vevea, 2017) to assess publication/inclusion bias. The bias tests include contoured-enhanced funnel plot with the trim-and-fill method (Borenstein et al., 2009; Duval, 2005; Peters et al., 2008), rank correlation test, Precision-effect test—precision-effect estimate with standard error (Stanley & Doucouliagos, 2014), Three-parameter selection model (Pustejovsky & Rodgers, 2019), Meta-regression test of publication type, Weight-function models (Coburn & Vevea, 2015), and *p*-uniform test (van Assen et al., 2015).

Confidence in cumulative estimate

Item 17. *Describe how the strength of the body of evidence will be assessed (e.g., GRADE).*

N/A

SECTION 4: Additional information

Deviations from the plan

We will use the deviation plan template (see ProtocolDeviations_meta-misinfo.xlsx) at the time of submission and will upload it to the OSF repository.

References

Arechar, A. A., Allen, J. N. L., Berinsky, A., Cole, R., Epstein, Z., Garimella, K., Gully, A., Lu, J. G., Ross, R. M., Stagnaro, M., Zhang, J., Pennycook, G., & Rand, D. (2022). *Understanding and reducing online misinformation across 16 countries on six continents*. PsyArXiv. https://doi.org/10.31234/OSF.IO/A9FRZ

Borenstein, M., Hedges, L., Higgins, J., & Rothstein, H. (2009). *Introduction to meta-analysis.* Wiley.

Calabrese, C., & Albarracin, D. (n.d.). *Unpublished data.*

Chan, M. S., Jones, C. R., Hall Jamieson, K., & Albarracín, D. (2017). Debunking: A meta-analysis of the psychological efficacy of messages countering misinformation. *Psychological Science*, *28*(11), 1531–1546. https://doi.org/http://dx.doi.org/10.1177/0956797617714579

Chan, M. S., Jones, C. R., Jamieson, K. H., Albarracin, D., Hall Jamieson, K., & Albarracín, D. (2017). Debunking: A meta-analysis of the psychological efficacy of messages countering misinformation. *Psychological Science*, *28*(11), 1531–1546. https://doi.org/10.1177/0956797617714579

Coburn, K. M., & Vevea, J. L. (2015). Publication bias as a function of study characteristics. *Psychological Methods*, *20*(3), 310–330. https://doi.org/10.1037/met0000046

Coburn, K. M., & Vevea, J. L. (2017). *weightr: Estimating weight-function models for publication bias* (R package version 1.1.2.).

Duval, S. (2005). The '"trim and fill"' method. In H. R. Rothstein, A. J. Sutton, & M. Borenstein (Eds.), *Publication bias in meta- analysis: Prevention, assessment, and adjustments* (pp. 127–144). John Wiley & Sons.

Ecker, U. K. H. (2022, February 23). *Correcting vaccine misinformation*. OSF. https://osf.io/dwyma/

Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., Kendeou, P., Vraga, E. K., & Amazeen, M. A. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology 2022 1:1*, *1*(1), 13–29. https://doi.org/10.1038/s44159-021-00006-y

Fisher, Z., & Tipton, E. (2015). *robumeta: An R-package for robust variance estimation in meta-analysis*. http://arxiv.org/abs/1503.02220

Gawronski, B., Brannon, S. M., & Ng, N. L. (2022). *Debunking misinformation about a causal link between vaccines and autism: Two preregistered tests of dual-process versus single-process predictions. (With conflicting results). Under review.*

Gesser-Edelsburg, A., Diamant, A., Hijazi, R., & Mesch, G. S. (2018). Correcting misinformation by health organizations during measles outbreaks: A controlled experiment. *PLOS ONE*, *13*(12), e0209505. https://doi.org/10.1371/JOURNAL.PONE.0209505

Greitemeyer, T., & Sagioglou, C. (2015). Does exonerating an accused researcher restore the researcher's credibility? *PLOS ONE*, *10*(5), e0126316. https://doi.org/10.1371/journal.pone.0126316

Hart, W., Albarracín, D., Eagly, A. H., Brechan, I., Lindberg, M. J., & Merrill, L. (2009). Feeling validated versus being correct: A meta-analysis of selective exposure to information. *Psychological Bulletin*, *135*(4), 555–588. https://doi.org/10.1037/a0015701

Hedges, L. v., Tipton, E., & Johnson, M. C. (2010). Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods*, *1*(1), 39–65. https://doi.org/10.1002/jrsm.5

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, *21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Higgins, J. P. T., Thompson, S. G., Deeks, J. J., & Altman, D. G. (2003). Measuring inconsistency in meta-analyses. *BMJ : British Medical Journal*, *327*(7414), 557–560. https://doi.org/10.1136/bmj.327.7414.557

Jahanbakhsh, F., Zhang, A. X., Berinsky, A. J., Pennycook, G., Rand, D. G., & Karger, D. R. (2021). Exploring lightweight interventions at posting time to reduce the sharing of misinformation on social media. *Proceedings of the ACM on Human-Computer Interaction*, *5*(CSCW1). https://doi.org/10.1145/3449092

Janmohamed, K., Walter, N., Sangngam, N., Hampsher, S., Nyhan, K., de Choudhury, M., & Kumar, N. (2022). Interventions to Mitigate Vaping Misinformation: A Meta-Analysis. *Journal of Health Communication*, *27*(2), 84–92. https://doi.org/10.1080/10810730.2022.2044941/FORMAT/EPUB

Koller, M. (1993). Rebutting accusations: When does it work, when does it fail? *European Journal of Social Psychology*, *23*(4), 373–389. https://doi.org/10.1002/ejsp.2420230405

Lewandowsky, S., Cook, J., Ecker, U. K. H., Albarracín, D., Amazeen, M. A., Kendeou, P., Lombardi, D., Newman, E. J., Pennycook, G., Porter, E., Rand, D. G., Rapp, D. N., Reifler, J., Roozenbeek, J., Schmid, P., Seifert, C. M., Sinatra, G. M., Swire-Thompson, B., van der Linden, S., … Zaragoza, M. S. (2020). *The Debunking Handbook 2020*. https://doi.org/10.17910/b7.1182

Lewandowsky, S., Ecker, U. K. H., Seifert, C. M., Schwarz, N., & Cook, J. (2012). Misinformation and its correction: Continued influence and successful debiasing. *Psychological Science in the Public Interest*, *13*(3), 106–131. https://doi.org/http://dx.doi.org/10.1177/1529100612451018

Lord, C. G., Ross, L., & Lepper, M. R. (1979). Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of Personality and Social Psychology*, *37*(11), 2098–2109.

McShane, B. B., Böckenholt, U., & Hansen, K. T. (2016). Adjusting for publication bias in meta-analysis: An evaluation of selection methods and some cautionary notes. *Perspectives on Psychological Science*, *11*(5), 730–749. https://doi.org/10.1177/1745691616662243

Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., Shekelle, P., Stewart, L. A., Estarli, M., Barrera, E. S. A., Martínez-Rodríguez, R., Baladia, E., Agüero, S. D., Camacho, S., Buhring, K., Herrero-López, A., Gil-González, D. M., Altman, D. G., Booth, A., … Whitlock, E. (2016). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Revista Espanola de Nutricion Humana y Dietetica*, *20*(2), 148–160. https://doi.org/10.1186/2046-4053-4-1/TABLES/4

Moreau, D., & Gamble, B. (2022). Conducting a Meta-Analysis in the Age of Open Science: Tools, Tips, and Practical Recommendations. *Psychological Methods*, *27*(3), 426–432. https://doi.org/10.1037/MET0000351

Mosleh, M., Martel, C., Eckles, D., & Rand, D. (2022). *Promoting engagement with social fact-checks online*. OSF Preprints. https://doi.org/10.31219/OSF.IO/RCKFY

Pennycook, G., Epstein, Z., Mosleh, M., Arechar, A. A., Eckles, D., & Rand, D. G. (2021). Shifting attention to accuracy can reduce misinformation online. *Nature 2021 592:7855*, *592*(7855), 590–595. https://doi.org/10.1038/s41586-021-03344-2

Pennycook, G., McPhetres, J., Zhang, Y., Lu, J. G., & Rand, D. G. (2020). Fighting COVID-19 misinformation on social media: Experimental evidence for a scalable accuracy-nudge intervention. *Psychological Science*, *31*(7), 770–780. https://doi.org/10.1177/0956797620939054

Pennycook, G., & Rand, D. G. (2019). Fighting misinformation on social media using crowdsourced judgments of news source quality. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(7), 2521–2526. https://doi.org/10.1073/PNAS.1806781116/SUPPL_FILE/PNAS.1806781116.SAPP.PDF

Peters, J. L., Sutton, A. J., Jones, D. R., Abrams, K. R., & Rushton, L. (2008). Contour-enhanced meta-analysis funnel plots help distinguish publication bias from other causes of asymmetry. *Journal of Clinical Epidemiology*, *61*(10), 991–996. https://doi.org/10.1016/j.jclinepi.2007.11.010

Pustejovsky, J. E., & Rodgers, M. A. (2019). Testing for funnel plot asymmetry of standardized mean differences. *Research Synthesis Methods*, *10*(1), 57–71. https://doi.org/10.1002/JRSM.1332

Seifert, C. M. (2002). The continued influence of misinformation in memory: What makes a correction effective? *Psychology of Learning and Motivation*, *41*, 265–292.

Sidik, K., & Jonkman, J. N. (2006). Robust variance estimation for random effects meta-analysis. *Computational Statistics and Data Analysis*, *50*(12), 3681–3701. https://doi.org/10.1016/j.csda.2005.07.019

Sirlin, N., Epstein, Z., Arechar, A. A., & Rand, D. G. (2021). Digital literacy is associated with more discerning accuracy judgments but not sharing intentions. *Harvard Kennedy School Misinformation Review*. https://doi.org/10.37016/MR-2020-83

Stanley, T. D., & Doucouliagos, H. (2014). Meta-regression approximations to reduce publication selection bias. *Research Synthesis Methods*, *5*(1), 60–78. https://doi.org/10.1002/jrsm.1095

van Assen, M. A. L. M., van Aert, R. C. M., & Wicherts, J. M. (2015). Meta-analysis using effect size distributions of only statistically significant studies. *Psychological Methods*, *20*(3), 293–309. https://doi.org/10.1037/MET0000025

van der Linden, S., Leiserowitz, A., Rosenthal, S., & Maibach, E. (2017). Inoculating the public against misinformation about climate change. *Global Challenges*, *1*(2), 1600008. https://doi.org/10.1002/gch2.201600008

Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*. https://doi.org/10.1103/PhysRevB.91.121108

Viechtbauer, W., & Cheung, M. W.-L. (2010). Outlier and influence diagnostics for meta-analysis. *Research Synthesis Methods*, *1*(2), 112–125. https://doi.org/10.1002/jrsm.11

Walter, N., Brooks, J. J., Saucier, C. J., & Suresh, S. (2021). Evaluating the Impact of Attempts to Correct Health Misinformation on Social Media: A Meta-Analysis. *Health Communication*, *36*(13), 1776–1784. https://doi.org/10.1080/10410236.2020.1794553/SUPPL_FILE/HHTH_A_1794553_SM1569.PDF

Walter, N., Cohen, J., Holbert, R. L., & Morag, Y. (2020). Fact-checking: A meta-analysis of what works and for whom. *Political Communication*, *37*(3), 350–375. http://search.ebscohost.com/login.aspx?direct=true&db=cms&AN=143116027

Walter, N., & Murphy, S. T. (2018). How to unring the bell: A meta-analytic approach to correction of misinformation. *Communication Monographs*, *85*(3), 423–441. https://doi.org/10.1080/03637751.2018.1467564

Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction: How powerful is it, why does it happen, and how to stop it?. *Communication Research*, *47*(2), 155–177. http://search.ebscohost.com/login.aspx?direct=true&db=cms&AN=142247666